

DEVESH RAVICHANDRAN

AI & Machine Learning Engineer | Agentic AI | Computer Vision | NLP

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OBJECTIVE

AI/ML engineer with hands-on experience in Agentic AI systems, Computer Vision, and NLP. Currently interning at CaratLane as a DevOps intern with a production mindset — I understand both how models are built and how they are shipped. Passionate about building end-to-end intelligent systems that solve real business problems.

EDUCATION

SRM Institute of Science and Technology

Chennai, India | 2022 – 2026

B.Tech — Computer Science, Specialisation in Artificial Intelligence & Machine Learning | CGPA: 7.85

- Worked on an AI Assistant for Dean (In-House Project)
- Participated in 2+ national-level hackathons representing SRM

PROFESSIONAL EXPERIENCE

DevOps Intern | CaratLane — A Tanishq Partnership

Jan 2025 – Present

- Designed and deployed a proof-of-concept for AWS Client VPN, enabling secure remote access to internal infrastructure and reducing dependency on on-premise VPN solutions
- Developed a User Access Review (UAR) tool to automate and streamline periodic access audits across internal systems — reducing manual review effort and improving compliance coverage
- Automated Freshservice ticket workflows using scripting and integration pipelines, reducing manual ticket handling time and improving IT operations efficiency
- Gained production-level exposure to CI/CD pipelines, containerisation (Docker), and infrastructure management, building a strong deployment and monitoring mindset for AI model delivery

AI/ML Intern | ONGC

Dec 2024 – Jan 2025

- Built a RAG-based PDF summarising chatbot and intelligent query bot for internal document retrieval
- Integrated LLM for contextual summarisation — improved document processing speed by 60% and query response by 95%
- Engineered a Streamlit UI with error handling, text cleaning, and FAISS-based semantic search

PROJECTS

Multi-Agent Student Learning System

Technologies: Python, LangChain, Streamlit, LLaMA 3.1, FAISS, Ollama Embeddings

- Designed a supervisor-router multi-agent architecture where specialised agents handle content retrieval, quiz generation, and student progress tracking
- Solved agent loop prevention using structured task ownership — each agent has defined scope and escalation path
- Built end-to-end pipeline from document ingestion to adaptive learning output using RAG + agentic orchestration
- Directly applicable to personalised customer journeys and intelligent recommendation flows in e-commerce

Social Media Management Agentic Platform (In Progress)

Technologies: Python, LangGraph, LLM APIs, Async Workflows

- Building a multi-agent platform with specialised agents for content ideation, scheduling, engagement monitoring, and real-time strategy pivots
- Engineering async agent workflows with state persistence — solving challenges not present in single-session agentic systems
- Demonstrates applied agentic architecture across multiple domains beyond education

3D Virtual Try-On Clothing Application

Technologies: Python, OpenCV, TensorFlow, Unity 3D, ARCore, React Native

- Built an AI-powered mobile app for virtual 3D clothing try-ons — core challenge was realistic garment warping across varied body poses
- Used pose estimation for accurate body mapping and thin-plate spline (TPS) transformations for garment fitting
- Integrated AR overlays for real-time garment visualisation — directly adjacent to jewellery AR try-on use cases
- Built as part of the clothing brand's website — runs locally; focused on getting the CV pipeline, warping, and pose estimation right before deployment

Attention-Guided Multi-Scale Feature Fusion for Skin Disease Classification

Technologies: Python, TensorFlow, Keras, Grad-CAM, LIME

- Designed CNN with attention gates for dermatological image classification — model learns to focus on diagnostically important regions
- Integrated Explainable AI (Grad-CAM, LIME) for visual interpretability — critical for trust in medical AI applications
- Achieved 91% classification accuracy through hyperparameter tuning and augmentation on limited medical datasets
- Attention gate approach directly applicable to fine-grained visual recognition in jewellery quality inspection

AI-Driven Intelligent Enterprise Assistant

Technologies: React.js, Express.js, Ollama AI, Gemini 2B, PostgreSQL, Elasticsearch, API Gateway

- Designed dual-chatbot system: Policy Bot (Ollama AI) and Employee Assistant with distinct routing logic
- Automated PDF-to-text conversion and image summarisation using Gemini 2B
- Integrated Elasticsearch and web scraping for contextual retrieval and real-time insights
- Built secure microservice communication pipeline using API Gateway — production-grade architecture

Frame Interpolation with Explainable AI

Technologies: Python, Deep Learning, XAI Techniques

- Working on video frame interpolation model that generates intermediate frames from sparse inputs
- Applying XAI to interpret which temporal features drive interpolation decisions — extending explainability beyond static image tasks

PUBLICATION

Comparative Analysis on BERT Models: ALBERT and RoBERTa

Tashin Khan, Thejal Ravichandran, Devesh Ravichandran

TECHNICAL SKILLS

ML / AI: Deep Learning, Computer Vision, NLP, Agentic AI, RAG, LLMs, Explainable AI, Transformers

Frameworks: TensorFlow, Keras, LangChain, LangGraph, Streamlit, Flask, React.js, Express.js, FAISS

Languages: Python, SQL, JavaScript, C, C++, Java, HTML, CSS

Databases: MySQL, PostgreSQL, NoSQL, Elasticsearch

Cloud / DevOps: AWS, Oracle Cloud, Docker, CI/CD pipelines

Other: Ollama, Gemini API, Meta-LLaMA 3.1, ARCore, Unity 3D, OpenCV

CERTIFICATIONS

- Oracle Cloud Infrastructure: Generative AI Professional
- AWS Academy Machine Learning Foundations
- AWS Academy Cloud Foundations
- AWS Academy Computer Networks

ACHIEVEMENTS

- Founder of a self-funded clothing brand — built an AI-powered virtual try-on system as part of the brand's website
- Secured 3rd Place in a National-Level Hackathon
- Mridangam — awarded in multiple elocution competitions